

PAUL MICKY D COSTA

SOFTWARE ENGINEER AND DATA SCIENTIST

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Extra Info: Reliable, Systematic, Detailed and logical

Communication & Languages: English: Fluent (C1) French: (A2)

SUMMARY

I bring **business value** — not just technical output. With a **combined IT and business background**, I **understand what users need**, speak their language, and **build solutions** around them. I work directly with end users, listen to their problems, and translate those into tools that are intuitive, purposeful, and adopted. I design with the business in mind: every dashboard, application, or automation I deliver is shaped by real use cases and measurable impact. I **pick up new tools fast**, **monitor and maintain** what I build, and proactively create solutions — for users, for colleagues, and for the organisation. My strength is sitting at the intersection of **data science, software engineering, and business understanding**: turning complex technical capabilities into clear, lasting value.

Acquired practical experience supporting **Quality, Health, Safety, Environment and Security (QHSES)** and **Risk Management** digital processes within a global engineering organisation. Produced **Product Architecture Reference Documents (PARDR)**, business cases, product roadmaps, training material, and executive presentations to support product adoption and decision-making. Worked in an international environment with stakeholders across multiple functions and geographies (**Business and Digital**).

EDUCATION

Master's in Computer Science and Data Science	2025–2026
ESILV - École Supérieure d'Ingénieurs Léonard de Vinci, France	
Bachelor of Science with Honours — First Class in Informatics	2024–2025
International Business Academy (IBA) (Certified by Coventry University), Kolding, Denmark	
Bachelor of Science in Computer Science and Engineering	2018–2023
Independent University, Bangladesh (IUB), Dhaka, Bangladesh	
General Certificate of Education — Advanced Subsidiary Level & Advanced Level (A-Levels)	2015–2017
International General Certificate of Secondary Education (IGCSE) / General Certificate of Education — Ordinary Level (O-Levels)	2014

PROFESSIONAL EXPERIENCE

Global QHSES Information Systems Officer (Intern) Technip Energies	30 May 2026 – 25 September 2026
Design and build integrating Intelex feedback records via API with the RMA risk database to automatically classify risks into four categories: risk identified & feedback recorded, risk identified & no feedback, unidentified risk with feedback occurred, and no risks defined (triggers AI-generated risk suggestions). Authored a PARDR (Product Architecture Reference Document / Platform Architecture Reference Document), user presentations, and businessvalue/roadmap documents targeting non-technical users.	

- Acquired practical experience supporting Quality, Health, Safety, Environment and Security (QHSES) and Risk Management digital processes within a global engineering organisation. Produced Product Architecture Reference Documents (PARD), business cases, product roadmaps, training material, and executive presentations to support product adoption and decision-making. Worked in an international environment with stakeholders across multiple functions and geographies (Business and Digiteam).
- Collaborated directly with **QHSES stakeholders** to **capture business requirements**, validate use cases, and iteratively design a solution aligned with operational risk management processes. **RTA (Real Time Alert) application** for the **Global Security team**: automated ingestion of live news sources, **immediate email alerts** to company addresses, 6hr AI digest generated, and **an interactive dashboard with geo-mapped incident locations**.
- Fully automated the **Global Quality Power BI dashboard** by replacing manual Excel feeds with a direct database connection (**SpotDB**), **eliminating manual data entry** and delivering a **live, organisation-wide Quality dashboard**.
- **Administered the TEN Alert platform** for the **Global Security Department**: managed user access and permissions, resolved user issues, maintained communication with platform users across the organisation, and **maintained ITIL-based monitoring** systems, tracking incidents and ensuring timely resolution of service issues.
- Developed a Hybrid RAG-powered **admin bot in Dataiku** to assist platform administrators with operational queries.
- Collaborated within the **QHSES Digital team** — **authoring BRs**, supporting colleagues, evaluating platforms, creating stakeholder presentations, and writing meeting minutes.
- Completed **T.ENET Training**, developing project outputs that were formally **reviewed and approved** by the organisation.

Front-End Developer | Tech Trioz Solutions

February 2023 – June 2023

- Designed and implemented UI/UX prototypes using Figma.
- Developed applications using Vue3 on Linux environments.
- Managed version control using GitHub.
- Produced essential project documentation (Gantt charts, flowcharts, technical reports).
- Key Projects: Developed "Shahajjo", an online services platform, and an employee tracking system for Grameenphone.

CERTIFICATIONS

- Advanced Designer Certificate (Dataiku)
- Core Designer Certificate (Dataiku)
- Developer Certificate (Dataiku)
- Generative AI Practitioner Certificate (Dataiku)
- MLOps Practitioner Certificate (Dataiku)
- ML Practitioner Certificate (Dataiku)
- T.ENET Training

DATA SCIENCE & SOFTWARE PROJECTS

Dataiku & BI Platform Projects (Technip Energies)

TEN — Intellex × RMA Risk Intelligence App - Dataiku, FastAPI, JavaScript, Pandas, Chart.js, Azure GPT-5.2

Built a web application that integrates **Intellex feedback** records (via REST API) with the **RMA risk** database to automatically classify risks across four categories: **risk identified & feedback recorded**; **risk identified & no feedback**; **unidentified risk with feedback occurred**; and **no risks defined** — in which case the system generates **AI-suggested risks**. Delivered as an 11-tab **dashboard** with 6 KPI cards, TF-IDF semantic matching engine, interactive charts (doughnut/bar/radar), real-time search, Excel export, and an Azure **GPT-5.2 AI assistant**. Authored a **PARD document**, user-facing presentations, business value documents, and product roadmaps with

non-technical users as the primary audience.

RealTimeAlerts - Automated Security Intelligence & News Monitoring Pipeline - Dataiku DSS, Python, Azure OpenAI (GPT-5.2), Apify, NewsAPI, FastAPI, Leaflet.js, Chart.js

Built an end-to-end automated threat-monitoring system on Dataiku combining NewsAPI/Apify ingestion (business-editable CSV config), **GPT-5.2-based** two-factor triage via Dataiku's LLM Mesh, URL-based deduplication, and tiered notifications (**SMTP email alerts**, scheduled reportlab PDF digests, Azure Blob Storage archival). Automated end-to-end via Dataiku Scenarios for unattended 24/7 operation, with a FastAPI/Leaflet.js/Chart.js dashboard for geo-mapping and live intelligence feeds, plus a cost-benefit architecture comparison presented to stakeholders.

Global Quality Dashboard - Power BI, SpotDB, TypeScript, Power Automate

Fully automated the organisation-wide **Global Quality dashboard** by migrating from manual Excel sheet feeds to a live **SpotDB** database connection, eliminating manual data entry. Delivering a real-time global Quality dashboard accessible across the organisation.

QHSES Project Portfolio 3D Network Visualization - Three.js, JavaScript, Dataiku

Designed and built an interactive **3D network visualization** deployed as a **Dataiku webapp**, mapping the QHSES Digital team's Quality, Health, Safety, Environment and Security project portfolio as a navigable **neuron-style graph** with animated signal pulses. Implemented click-through navigation linking each node to **its live dashboard**, SharePoint documentation, and Dataiku flow, using custom 3D rendering (camera controls, raycasting, dynamic curve-based connectors) rather than off-the-shelf charting libraries.

Audit Findings Executive Summary Automation - Dataiku DSS, Flask, Azure OpenAI (GPT-4o), Intellex REST API (OData v2)

Built an AI solution on Dataiku that retrieves audit records from Intellex via its OData v2 API and uses GPT-4o to generate structured executive summaries (risk assessment, key themes, critical risks, recommendations). Delivered as a self-service **Flask web app** and a batch-processing recipe with per-row error handling, backed by a full Business Requirements Document for stakeholder sign-off.

TripAdvisor Hotel Review Sentiment Analysis & GenAI Insights - Dataiku DSS, GenAI Recipes, LLM Mesh (GPT-4o), Visual AI Agents

Built an end-to-end GenAI pipeline in Dataiku using Classify/Summarize/Prompt recipes for sentiment analysis, summarization, and multilingual translation of hotel reviews, plus a custom Agent Tool for keyword-based review search and a **Visual AI Agent** enabling natural-language querying of the review dataset.

Distributed Systems & Real-Time Data Pipeline

Real-Time Weather Data Pipeline - Kafka, Spark, Hadoop, Azure, InfluxDB, Grafana

Designed and implemented a multi-node, scalable data pipeline on **Azure Linux VMs (Master/Worker)** for real-time data ingestion. Utilized **Apache Kafka for API ingestion** (OpenAPI 3.0), **Hadoop/PySpark** (YARN) for distributed processing, and stored time-series data in InfluxDB (PostgreSQL for structured data). Visualized metrics in Grafana.

RAG & Local LLM Systems

Medical RAG Chatbot - Python, ChromaDB, Sentence-Transformers, Ollama (Mistral Nemo), HuggingFace

Built a fully local **Retrieval-Augmented Generation (RAG) system** that answers medical questions by retrieving semantically relevant context from 33,955 medical Q&A pairs (medical_meadow_medical_flashcards dataset). Embeddings are generated with **all-MiniLM-L6-v2** and indexed in ChromaDB; at query time the top-5 most similar records are injected as context into **Mistral Nemo (served via Ollama)** which streams a grounded answer. The pipeline runs entirely on-device with no external API calls, making it suitable for privacy-sensitive medical data.

Machine Learning & Forecasting

Criteo Campaign Analysis (CTR Prediction and Optimization) - Machine Learning Project

Machine Learning project focused on predicting and optimizing Click-Through Rate (CTR) of Criteo advertisements using Random Forest models.

French Government Expenditure Analysis (1995-2030 Forecast) - Data Analysis & Forecasting

Data-driven analysis and forecasting of French public expenditure using regression models and visualization tools to predict trends up to 2030.

OSINT & Security Intelligence

Global Risk Watch (globalriskwatch.paulmickydcosta.com) - Real-Time Geopolitical Risk Monitoring Platform - Node.js, Express, D3.js, Cloudflare (DNS/Tunnel), NSSM

Solo-built, end-to-end full-stack platform aggregating live global news data and public security records into an interactive world map with country-level risk scoring, cross-border relationship graphs, and role-based data layers. Built a live data pipeline ingesting **80,000+ daily events** from **GDELT's public API**, computing a weighted risk score (conflict ratio, sentiment, event volume) and cross-border relationship graphs per country in near real-time. Integrated **INTERPOL's public Red Notices API** — including working around its bot-protection layer — to build a wanted-persons data layer with on-demand detail fetching and local caching, and designed a server-side text-classification pipeline (keyword search plus local verification against full record text) to derive two additional data categories from unstructured charge data while minimising false positives. Built an interactive **D3.js** visualisation layer: a projected world map with dynamic risk bubbles, curved relationship arcs, geocoded event markers, and live filtering across multiple data dimensions. Implemented authentication and security hardening — session-based auth with timing-safe credential comparison and brute-force lockout, a strict Content-Security-Policy, and remediation of a stored-XSS vulnerability across 20+ render paths — then deployed to production on self-managed infrastructure: migrated DNS to **Cloudflare**, configured a **Cloudflare Tunnel** for public HTTPS access with zero open inbound ports, and set up auto-restarting Windows services for continuous uptime.

France Situational Awareness Monitor (monitor.paulmickydcosta.com) - Node.js, Express, D3.js, Chart.js, Anthropic Claude API, RSS/Atom/XML, HTML parsing, JavaScript

Built a self-hosted full-stack OSINT threat-monitoring dashboard aggregating real-time news, video, and social media content from **27+ open sources** (RSS/Atom feeds, YouTube channels, public Telegram channels). Designed a **bilingual (EN/FR) rule-based classification engine** (~45 threat signals, ~25 France-relevance signals, ~120-location gazetteer) to auto-tag articles by category, threat level, and geographic origin. Integrated the **Anthropic Claude API** to generate structured AI threat briefings (threat rationale, key indicators, response actions, urgency assessment) via an async worker queue with per-item caching to control API cost. Built interactive visualisations with **D3.js** (world map with geo-projected markers, force-directed co-occurrence graph) and **Chart.js** (category/timeline/source dashboards), served via a REST API to a vanilla-JS SPA with live filtering, search, and auto-refresh. Implemented graceful degradation and retry logic for third-party API failures.

Application Development

Buddy — Personal AI Avatar (buddy.paulmickydcosta.com) - Conversational AI, AR, Meta Quest 3 Pro, Web

Built and deployed a personal AI avatar accessible at buddy.paulmickydcosta.com. Visitors can hold a live conversation with the avatar and view it in real time. Extended the experience into Augmented Reality using Meta Quest 3 Pro — bringing the avatar to life in physical space for a realistic, immersive interaction that elevates user experience beyond a standard screen interface. Demos published as YouTube Shorts:

youtube.com/@PaulDCosta-l7g/shorts.

TECHNICAL SKILLS

Advanced Analytics & AI

Advanced Deep Learning: Natural Language Processing (NLP), Computer Vision.

Graph Systems & Mining: Expertise in graph database concepts and network analysis.

Immersive Data Visualization / Visual Analytics: Creating interactive, exploratory, and immersive visualization solutions.

Web Scraping & Data Processing

Big Data & Distributed Process: **Apache Spark, PySpark (YARN), Hadoop**.

Scraping & Processing Tools: **Python, Scala, BeautifulSoup**.

Expertise: **Designing scalable pipelines for real-time API ingestion (OpenAPI 3.0) and complex web scraping**.

Databases, BI & MLOps

Machine Learning Operations (MLOps): **CI/CD pipelines (Jenkins, Docker, SonarQube) for automated, reproducible deployments**.

Business Intelligence (BI): **Power BI, Grafana (real-time dashboarding)**.

Data Storage: **PostgreSQL (PgAdmin), InfluxDB (time-series), MongoDB, Superbase, SpotDB, DataGalaxy.**

Cloud & Environment: **Microsoft Azure, AWS (Linux VMs), VM Linux Ubuntu/Kali, Jupyter/Colab, Dataiku DSS.**

Security, AI & Sustainability Skills

Sustainability Measurement: assessing and reporting sustainability/QHSE performance indicators.

Web Security: **security cookies, authentication & password management, ethical hacking.**

Generative AI: **Retrieval-Augmented Generation (RAG), API key management, training and developing new ML/AI models, LLM Mesh & Visual AI Agents (Dataiku).**

Security & Software Tools

Microsoft Threat Modeling Tool, OWASP ZAP, SonarQube.

Low-Code & Automation Platforms

Dataiku DSS: **end-to-end data pipelines, GenAI recipes, LLM Mesh, Visual AI Agents, Scenarios, and Dataiku webapps (Flask/FastApi/Three.js/Chart.js).**

Power Automate: **workflow automation and dashboard refresh/alert triggers.**

Communication & Documentation

Creating **presentations (PowerPoint)** for project updates and stakeholder communication.

Writing meeting minutes and Business Requirements (BR) documentation.

Core Engineering

Languages: Python, C#, C++, Java, Bash, Scala, SQL, flux.

Web/Front-End: HTML, CSS, JavaScript, Vue.js, React, Node.js.

Tools: VSCode, Cursor, Git, GitHub, Cisco, **InteleX, SharePoint Online, Microsoft 365, Copilot, TopoOne, TEN Alert, GardaWorld Travel Security, RMA, DragonFly.**

Low-Code/Data Platforms: Dataiku DSS, Power Automate.

Big Data & Distributed: Apache Spark, PySpark (YARN), Hadoop, Apache Kafka

Cloud & Infrastructure: Microsoft Azure, AWS (Linux VMs), Docker, VM Linux Ubuntu, Kali Linux

Data Storage & BI: **PostgreSQL (PgAdmin), InfluxDB (time-series), MongoDB, Superbase, SpotDB, DataGalaxy, Power BI, Grafana**

MLOps & CI/CD: **Jenkins, SonarQube, n8n, MLflow, Docker Compose, PyTest.**

Advanced Analytics & AI: Advanced Deep Learning, Natural Language Processing (NLP), Computer Vision, Graph Systems & Mining, Immersive Data Visualization, Visual Analytics, Beautiful Soup, Generative AI (LLMs)

Data Science Platforms: **Jupyter Notebooks, Google Colab**